



NAME: ABDULLAH SOHAIL

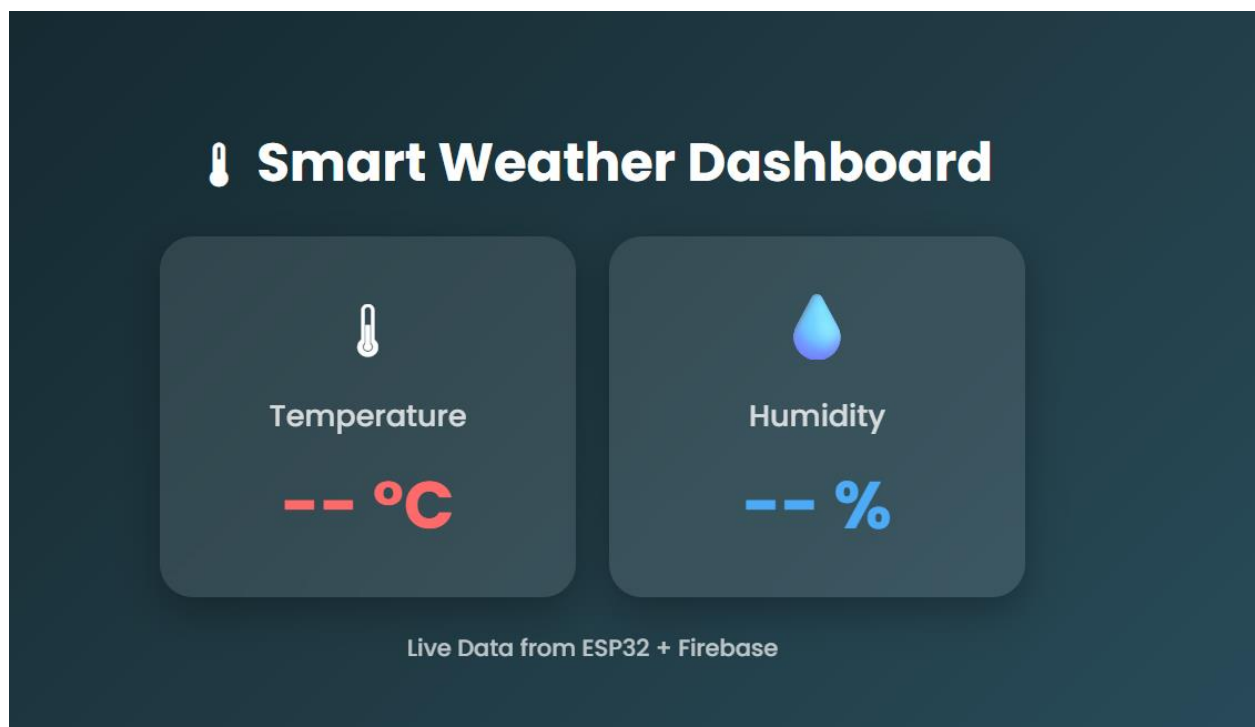
SAP: 70147165

SEC: BSIET (I)

SUBMITTED TO: DR. SYED MUHAMMAD HAMEDOON

ASSIGNMENT # 5:-

TEMPERATURE SENSOR:



```

1  #include <WiFi.h>
2  #include <HttpClient.h>
3  #include "DHT.h"
4
5  // ===== DHT SETUP =====
6  #define DHTPIN 4
7  #define DHTTYPE DHT22
8
9  DHT dht(DHTPIN, DHTTYPE);
10
11 // ===== WIFI =====
12 const char* ssid = "Wokwi-GUEST";
13 const char* password = "";
14
15 // ===== FIREBASE URL (IMPORTANT: .json) =====
16 String firebaseURL = "https://task4-f61fc-default-rtdb.firebaseio.com/se
17
18 void setup() {
19     Serial.begin(115200);
20     dht.begin();
21
22     // Connect WiFi
23     WiFi.begin(ssid, password);
24     Serial.print("Connecting to WiFi");

```

```

22     // Connect WiFi
23     WiFi.begin(ssid, password);
24     Serial.print("Connecting to WiFi");
25
26     while (WiFi.status() != WL_CONNECTED) {
27         delay(500);
28         Serial.print(".");
29     }
30
31     Serial.println("\n✅ Connected to WiFi!");
32 }
33
34 void loop() {
35
36     // ===== READ SENSOR =====
37     float temperature = dht.readTemperature();
38     float humidity = dht.readHumidity();
39
40     // ===== CHECK ERROR =====
41     if (isnan(temperature) || isnan(humidity)) {
42         Serial.println("❌ Failed to read from DHT sensor!");
43         delay(2000);
44         return;
45     }

```

```

47 // ===== PRINT VALUES =====
48 Serial.print("🌡 Temperature: ");
49 Serial.print(temperature);
50 Serial.println(" °C");
51
52 Serial.print("💧 Humidity: ");
53 Serial.print(humidity);
54 Serial.println(" %");
55
56 // ===== SEND TO FIREBASE =====
57 if (WiFi.status() == WL_CONNECTED) {
58     HTTPClient http;
59
60     http.begin(firebaseURL);
61     http.addHeader("Content-Type", "application/json");
62
63     // ===== JSON DATA =====
64     String jsonData = "{";
65     jsonData += "\"temperature\":\"" + String(temperature) + ",";
66     jsonData += "\"humidity\":\"" + String(humidity);
67     jsonData += "}";

```

```

69 // ===== SEND DATA =====
70 int httpResponseCode = http.PUT(jsonData);
71
72 Serial.print("📡 Firebase Response Code: ");
73 Serial.println(httpResponseCode);
74
75 // ===== PRINT RESPONSE BODY =====
76 String response = http.getString();
77 Serial.println("📄 Response:");
78 Serial.println(response);
79
80 http.end();
81 }
82 else {
83     Serial.println("❌ WiFi Disconnected!");
84 }
85
86 delay(5000); // send every 5 seconds
87 }

```

🔧 Smart Dashboard

Temperature

24 °C

Humidity

40 %

✅ Connected & Reading Data

<https://task4-f61fc-default-rtdb.firebaseio.com>

<https://task4-f61fc-default-rtdb.firebaseio.com/>

▼ — sensor

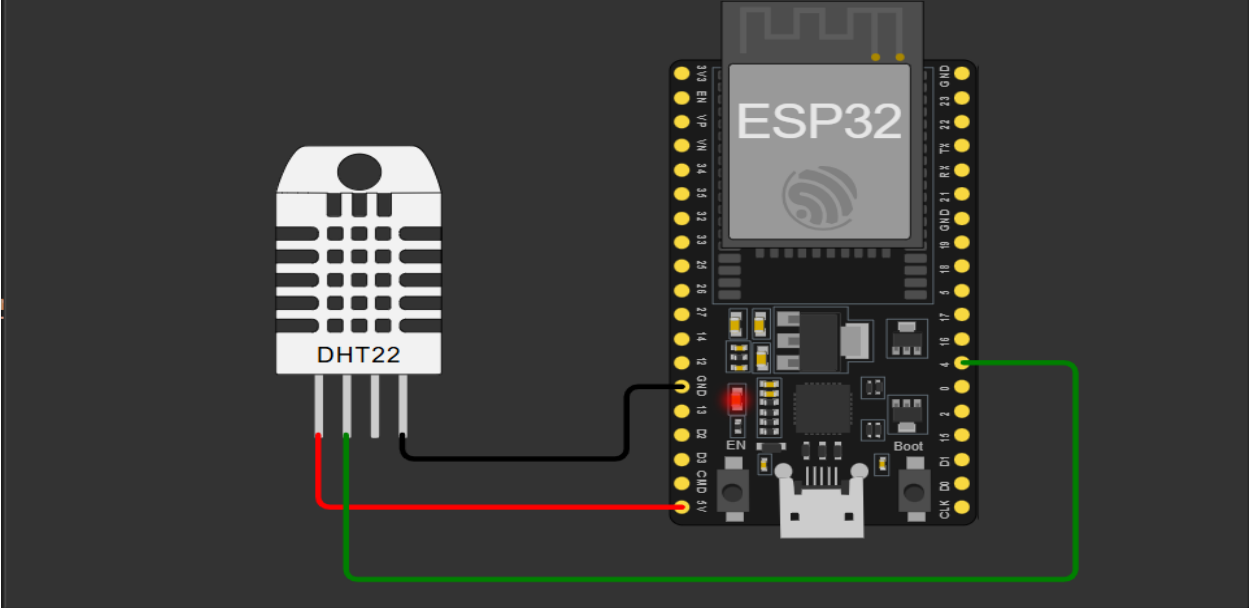
humidity: 40

temperature: 24

Docs

Simulation

00:39.315 86%



Response: 200

load:0x40080400,len:2972
entry 0x400805dc
Connecting.....
Connected!
Temperature: 24.00
Humidity: 40.00
Response: 200